

Carlos Coronel • Steven Morris

Database Systems

Design, Implementation, & Management

1/15/2024

Database System Design, Implementation And Management

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National Economics University
Faculty of Mathematical Economics

Database System Design, Implementation And
Management Course – Entity Relationship (ER)
Modeling

- Instructor: Duc Minh Vu (MFE)
- Email: minhvd [at] neu.edu.vn

1/15/2024

Database System Design, Implementation And Management

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Learning Objectives

1. Identify the main characteristics of entity relationship components
2. Describe how relationships between entities are defined, refined, and incorporated into the database design process
3. Explain how ERD components affect database design and implementation
4. Describe how real-world database design often requires there conciliation of conflicting goals

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Introduction

- ❑ Good decisions require good information derived from raw facts
- ❑ Data is managed most efficiently when stored in a database
- ❑ Databases evolved from computer file systems
- ❑ Understanding file system characteristics is important

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4-1 The Entity Relationship Model

4-1a Entities:

- an entity actually refers to the *entity set* and not to a single entity occurrence.
- an *entity* in the ERM corresponds to a table—not to a row
- The ERM refers to a table row as an *entity instance* or *entity occurrence*

4-1b Attributes

- Attributes are characteristics of entities.
- **Required and Optional Attributes**
 - ✓ A **required attribute** is an attribute that must have a value
 - ✓ An **optional attribute** is an attribute that does not require a value
- A **domain** is the set of possible values for a given attribute

required attribute

In ER modeling, an attribute that must have a value. In other words, it cannot be left empty.

optional attribute

In ER modeling, an attribute that does not require a value; therefore, it can be left empty.

domain

The possible set of values for a given attribute.

4-1 The Entity Relationship Model

4-1b Attributes

Identifiers (Primary Keys)

- ✓ The ERM uses **identifiers**—one or more attributes that uniquely identify each entity instance.
- ✓ Identifiers are underlined in the ERD
- ✓ Key attributes are underlined in a frequently used shorthand notation for the table structure, called a **relational schema**, that uses the following format

TABLE NAME (KEY_ATTRIBUTE 1, ATTRIBUTE 2, ATTRIBUTE 3, ... ATTRIBUTE K)

For example, a CAR entity may be represented by

CAR (CAR_VIN, MOD_CODE, CAR_YEAR, CAR_COLOR)

Each car is identified by a unique vehicle identification number, or CAR_VIN.

identifier

One or more attributes that uniquely identify each entity instance.

relational schema

The organization of a relational database as described by the database administrator.

4-1 The Entity Relationship Model

4-1b Attributes

➤ Composite Identifiers

- ✓ **composite identifier**, a primary key composed of more than one attribute

✓

composite identifier

In ER modeling, a key composed of more than one attribute.

CLASS (CRS_CODE, CLASS_SECTION, CLASS_TIME, ROOM_CODE, PROF_NUM)

Note that *both* key attributes are underlined in the entity notation.

4-1 The Entity Relationship Model

4-1b Attributes

➤ Composite and Simple Attributes

- ✓ A **composite attribute** is an attribute that can be further subdivided to yield additional attributes.
- ✓ A **simple attribute** is an attribute that cannot be subdivided.

composite attribute

An attribute that can be further subdivided to yield additional attributes. For example, a phone number such as 615-898-2368 may be divided into an area code (615), an exchange number (898), and a four-digit code (2368). Compare to *simple attribute*.

simple attribute

An attribute that cannot be subdivided into meaningful components. Compare to *composite attribute*.

4-1 The Entity Relationship Model

4-1b Attributes

- A **single-valued attribute** is an attribute that can have only a single value.
 - ✓ E.g. Social Security Number
 - ✓ A single-valued attribute is not necessarily a simple attribute
- **Multivalued attributes** are attributes that can have many values.
 - ✓ Degrees, phone numbers

single-valued attribute

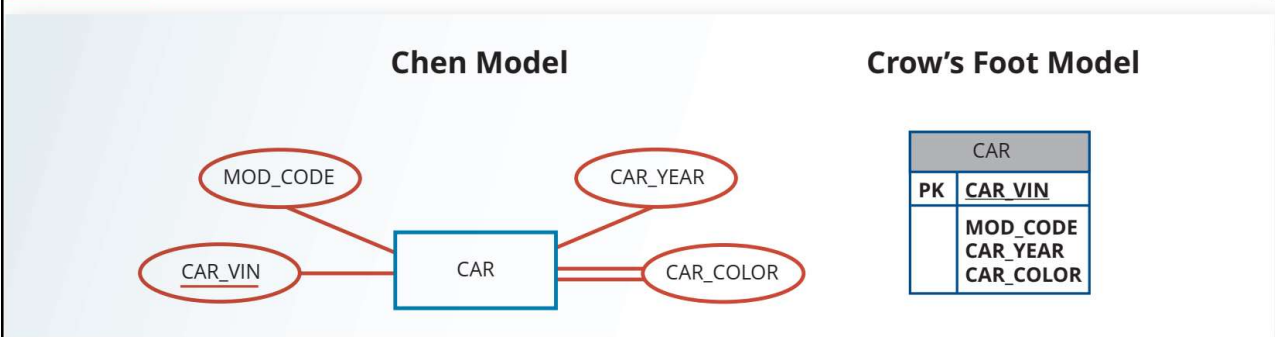
An attribute that can have only one value.

multivalued attribute

An attribute that can have many values for a single entity occurrence. For example, an EMP_DEGREE attribute might store the string "BBA, MBA, PHD" to indicate three different degrees held.

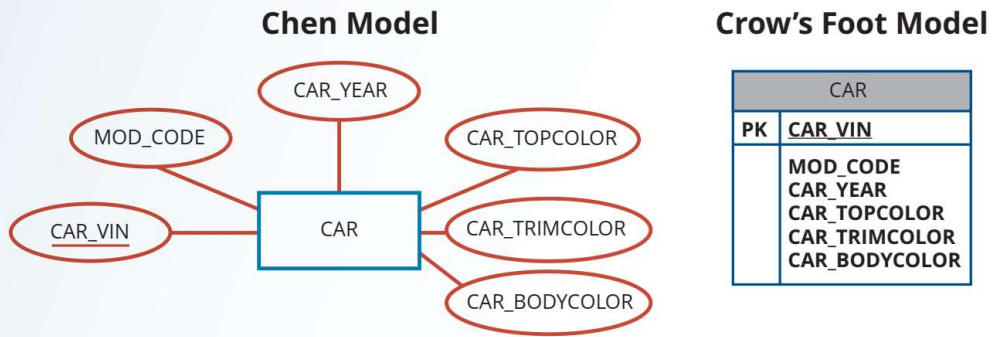
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Figure 4.3 A Multivalued Attribute in An Entity



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Figure 4.4 Splitting the Multivalued Attribute into New Attributes



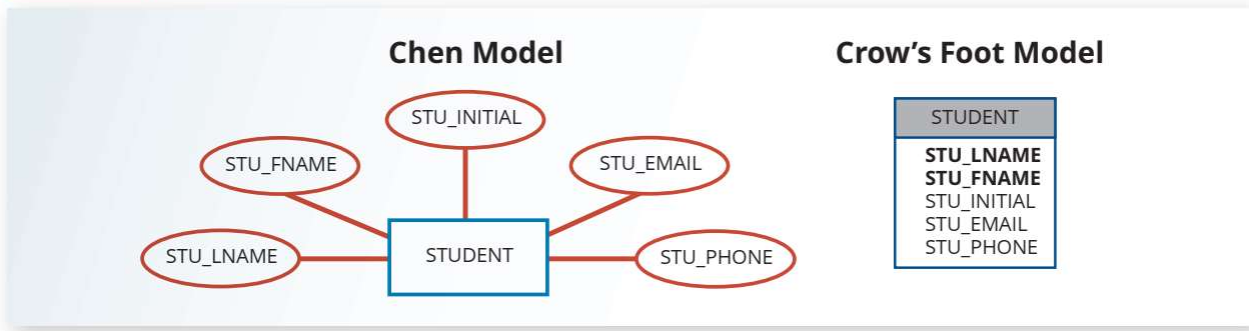
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Figure 4.5 A New Entity Set Composed of a Multivalued Attribute's Components



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Figure 4.1 The Attributes of the STUDENT Entity: Chen and Crow's Foot



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4-1 The Entity Relationship Model

4-1b Attributes

- A **derived attribute** is an attribute whose value is calculated (derived) from other attributes.

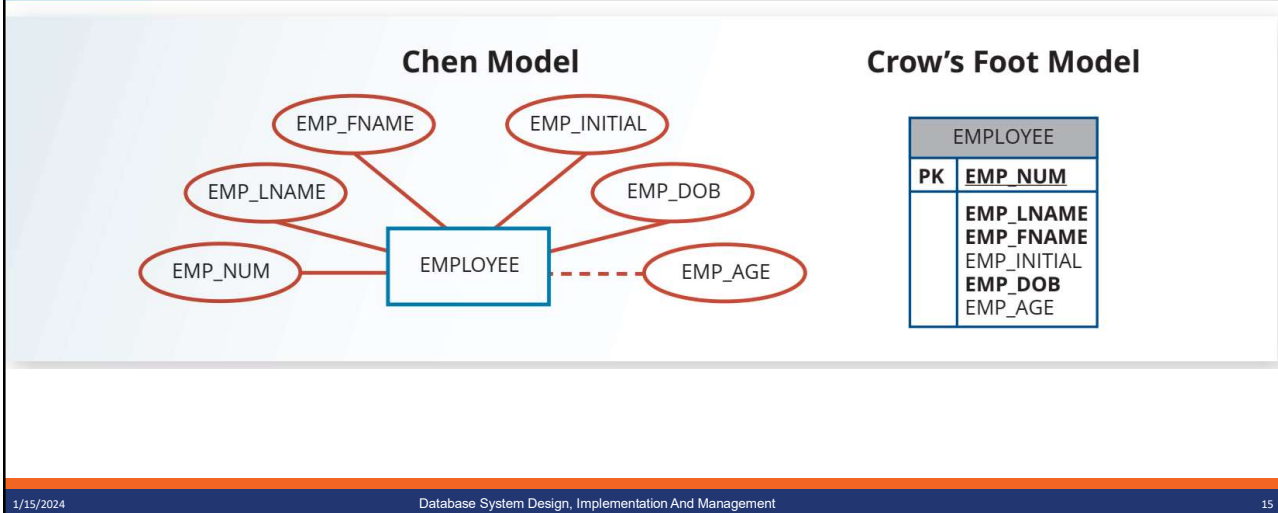
✓ Age = DATE() - DOB

derived attribute

An attribute that does not physically exist within the entity and is derived via an algorithm. For example, the Age attribute might be derived by subtracting the birth date from the current date.

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Figure 4.6 Depiction of a Derived Attribute



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Table 4.2 Advantages and Disadvantages of Storing Derived Attributes

	Derived Attribute	
	Stored	Not Stored
Advantage	- Saves CPU processing cycles - Saves data access time - Data value is readily available - Can be used to keep track of historical data	- Saves storage space - Computation always yields current value
Disadvantage	Requires constant maintenance to ensure derived value is current, especially if any values used in the calculation change	- Uses CPU processing cycles - Increases data access time - Adds coding complexity to queries

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4-1 The Entity Relationship Model

4-1c Relationships

- A relationship is an association between entities
- The entities that participate in a relationship are also known as **participants**, and each relationship is identified by a name that describes the relationship
- Relationships between entities always operate in both directions

A CUSTOMER may generate many INVOICES.
Each INVOICE is generated by one CUSTOMER.

4-1 The Entity Relationship Model

4-1d Connectivity and Cardinality

- Entity relationships may be classified as one-to-one, one-to-many, or many-to-many
- The term **connectivity** is used to describe the relationship classification.
- **Cardinality** expresses the minimum and maximum number of entity occurrences associated with one occurrence of the related entity

Figure 4.7 Connectivity and Cardinality in an ERD

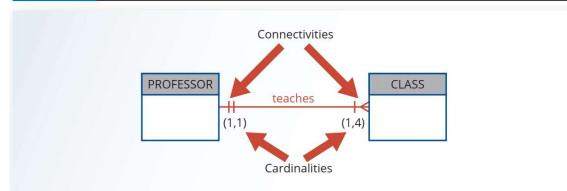
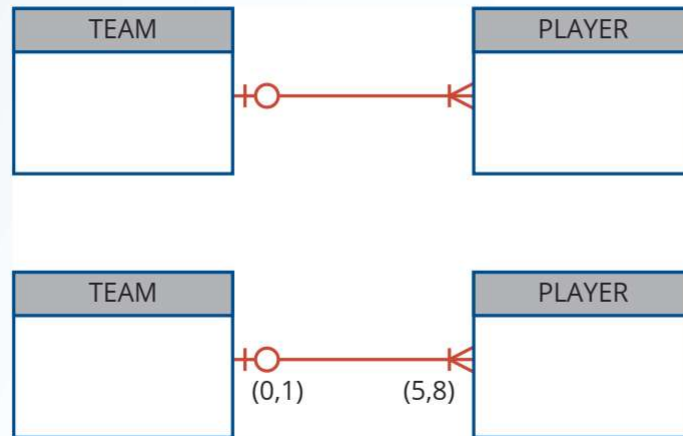


Figure 4.8 Two Representations of TEAM and PLAYER



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4-1 The Entity Relationship Model

4-1e Existence Dependence

- An entity is said to be **existence-dependent** if it can exist in the database only when it is associated with another related entity occurrence.
- If an entity can exist apart from all of its related entities, then it is **existence-independent**, and it is referred to as a **strong entity** or **regular entity**

existence-dependent

A property of an entity whose existence depends on one or more other entities. In such an environment, the existence-independent table must be created and loaded first because the existence-dependent key cannot reference a table that does not yet exist.

existence-independent

A property of an entity that can exist apart from one or more related entities. Such a table must be created first when referencing an existence-dependent table.

strong entity

An entity that is existence-independent, that is, it can exist apart from all of its related entities.

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4-1 The Entity Relationship Model

4-1f Relationship Strength

- A **weak relationship**, also known as a **non-identifying relationship**, exists if the primary key of the related entity does not contain a primary key component of the parent entity.
- A **strong (identifying) relationship** exists when the primary key of the related entity contains a primary key component of the parent entity.

weak (non-identifying) relationship

A relationship in which the primary key of the related entity does not contain a primary key component of the parent entity.

strong (identifying) relationship

A relationship that occurs when two entities are existence-dependent; from a database design perspective, this relationship exists whenever the primary key of the related entity contains the primary key of the parent entity.

Figure 4.9 A Weak (Non-Identifying) Relationship Between COURSE and CLASS

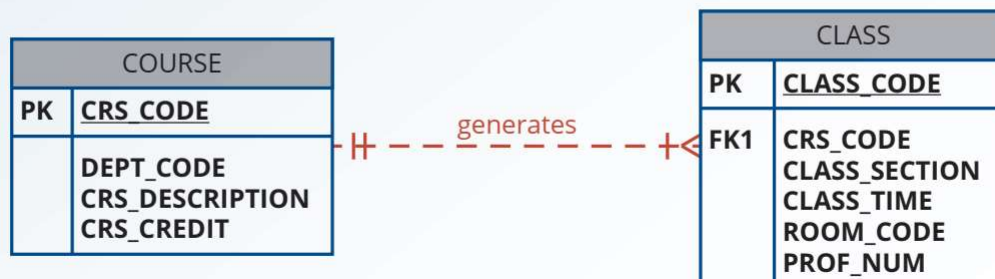
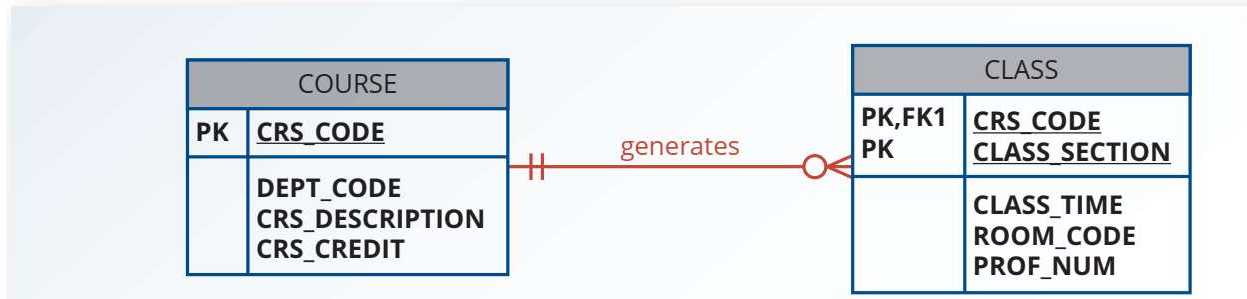


Figure 4.10 A Strong (Identifying) Relationship Between COURSE and CLASS



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4-1 The Entity Relationship Model

4-1g Weak Entities

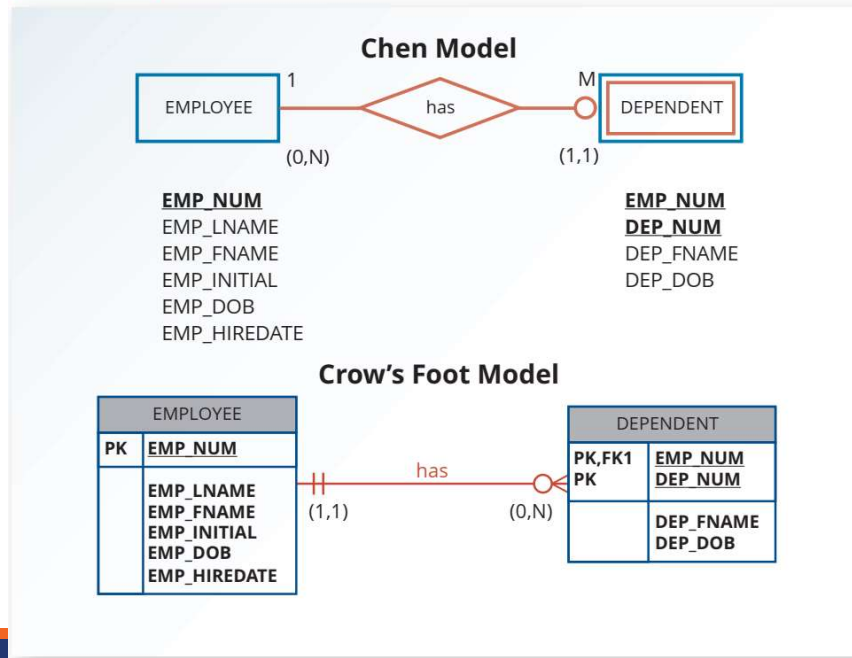
- A **weak entity** is one that meets two conditions
 - ✓ 1. The entity is existence-dependent; it cannot exist without the entity with which it has a relationship.
 - 2. The entity has a primary key that is partially or totally derived from the parent entity in the relationship.

weak entity

An entity that displays existence dependence and inherits the primary key of its parent entity. For example, a **DEPENDENT** requires the existence of an **EMPLOYEE**.

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Figure 4.11 A Weak Entity in an ERD



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4-1 The Entity Relationship Model

4-1h Relationship Participation

- **Optional participation** means that one entity occurrence does not *require* a corresponding entity occurrence in a particular relationship
 - ✓ COURSE vs CLASS
 - ✓ the CLASS entity is considered to be *optional* to the COURSE entity
- **Mandatory participation** means that one entity occurrence *requires* a corresponding entity occurrence in a particular relationship.

optional participation

In ER modeling, a condition in which one entity occurrence does not require a corresponding entity occurrence in a particular relationship.

mandatory participation

A relationship in which one entity occurrence must have a corresponding occurrence in another entity. For example, an EMPLOYEE works in a DIVISION. (A person cannot be an employee without being assigned to a company's division.)

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Table 4.3 Crow's Foot Symbols

Symbol	Cardinality	Comment
	(0,N)	Zero or many; the "many" side is optional.
	(1,N)	One or many; the "many" side is mandatory.
	(1,1)	One and only one; the "1" side is mandatory.
	(0,1)	Zero or one; the "1" side is optional.

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Figure 4.13 An Optional CLASS Entity in the Relationship "PROFESSOR teaches CLASS"



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Figure 4.14 CLASS Is Optional to COURSE

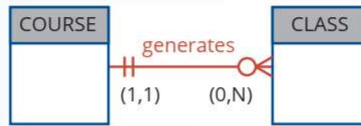
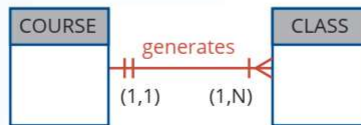


Figure 4.15 COURSE and CLASS in a Mandatory Relationship



4-1 The Entity Relationship Model

4-1i Relationship Degree

- A **relationship degree** indicates the number of entities or participants associated with a relationship.
- A **unary relationship** exists when an association is maintained within a single entity.
- A **binary relationship** exists when two entities are associated.
- A **ternary** exists when three entities are associated.
- **Recursive relationship**: an entity has a relationship with itself.

relationship degree

The number of entities or participants associated with a relationship. A relationship degree can be unary, binary, ternary, or higher.

unary relationship

An ER term used to describe an association *within* an entity. For example, an EMPLOYEE might manage another EMPLOYEE.

binary relationship

An ER term for an association (relationship) between two entities. For example, PROFESSOR teaches CLASS.

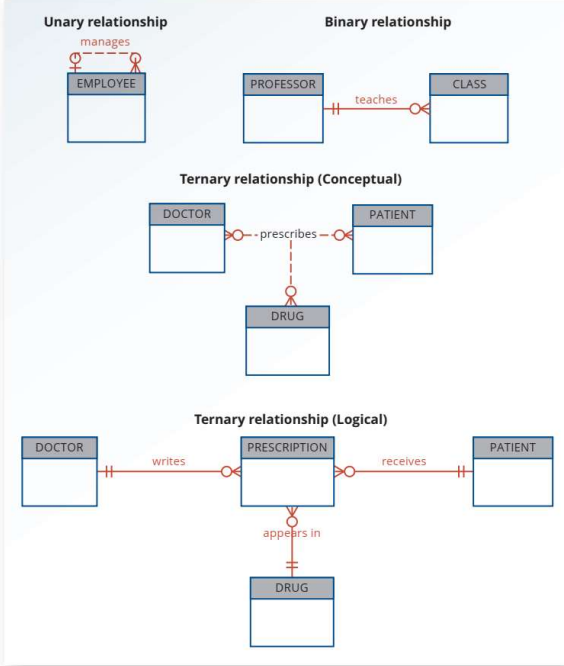
ternary relationship

An ER term used to describe an association (relationship) between three entities. For example, a DOCTOR prescribes a DRUG for a PATIENT.

recursive relationship

A relationship found within a single entity type. For example, an EMPLOYEE is married to an EMPLOYEE or a PART is a component of another PART.

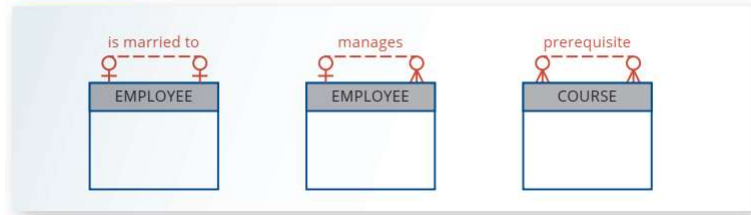
Figure 4.16 Three Types of Relationship Degree



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Figure 4.18 An ER Representation of Recursive Relationships



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Figure 4.19 The 1:1 Recursive Relationship “EMPLOYEE is married to EMPLOYEE”

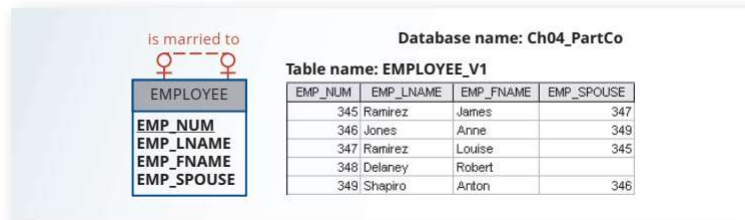


Figure 4.20 Implementation of the 1:M Recursive Relationship “EMPLOYEE manages EMPLOYEE”

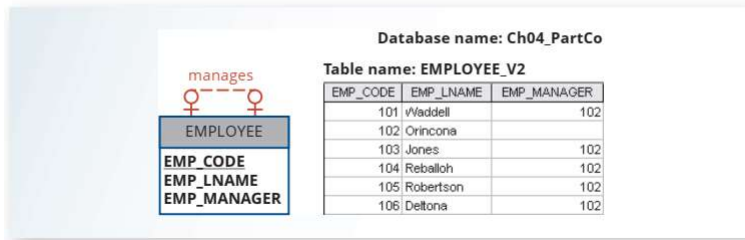
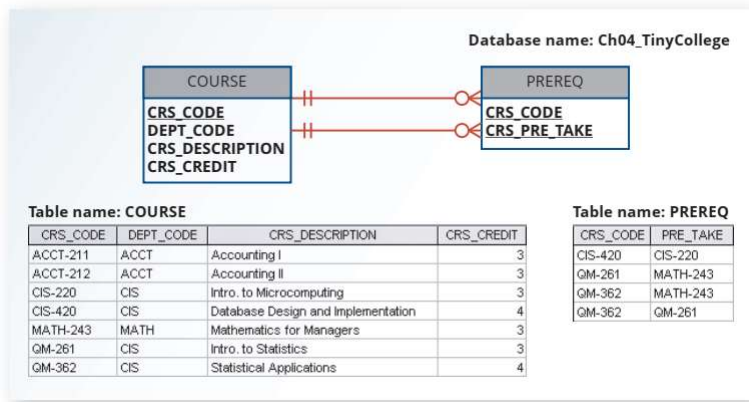




Figure 4.21 Implementation of the M:N Recursive Relationship
"COURSE requires COURSE"



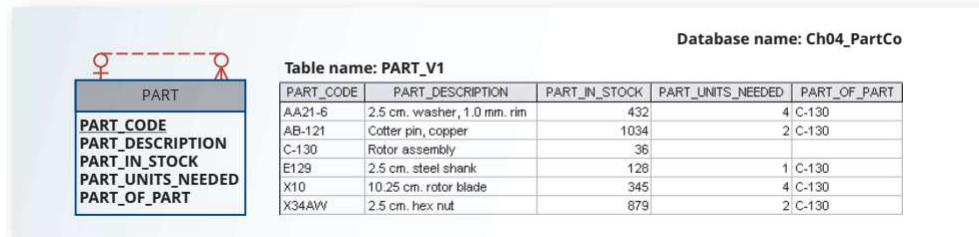
□ two 1:M relationships between COURSE and PREREQ

□ one copy of CRS_CODE was renamed to CRS_PRE TAKE

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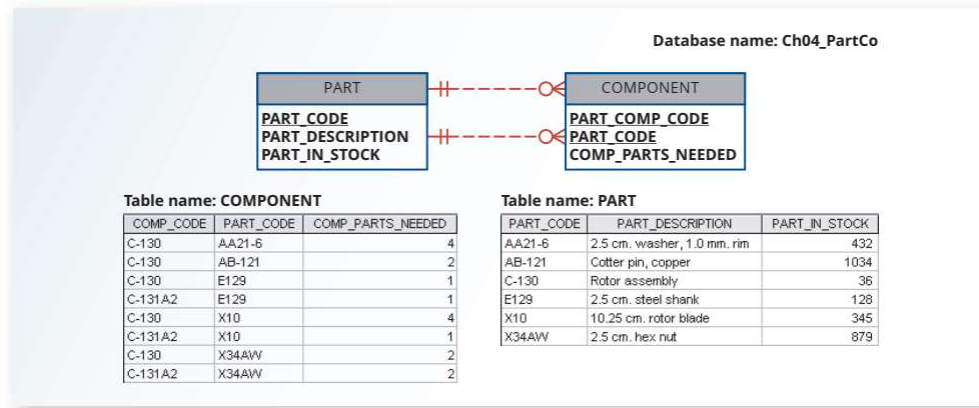


Figure 4.22 Another Unary Relationship: "PART Contains PART"



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Figure 4.23 The Implementation of the M:N Recursive Relationship “PART Contains PART”



4-2 Developing an ER Diagram

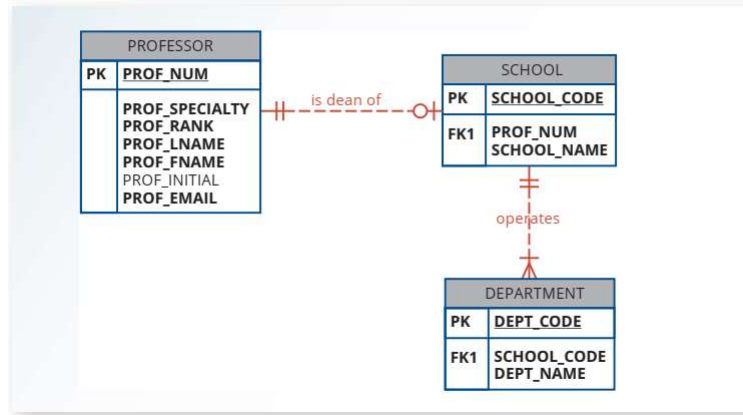
4-2 Developing an ER Diagram

- ❑ Create a detailed narrative of the organization's description of operations.
- ❑ Identify the business rules based on the description of operations.
- ❑ Identify the main entities and relationships from the business rules.
- ❑ Develop the initial ERD.
- ❑ Identify the attributes and primary keys that adequately describe the entities.
- ❑ Revise and review the ERD

Tiny College ERD

- ❑ Tiny College (TC) is divided into several schools. Each school is administered by a dean who is a professor. Each professor can be the dean of only one school, and a professor is not required to be the dean of any school.
 - a 1:1 relationship exists between PROFESSOR and SCHOOL
- ❑ Each school comprises several departments. On the other hand, each department belongs to only a single school.

Figure 4.27 The First Tiny College ERD Segment

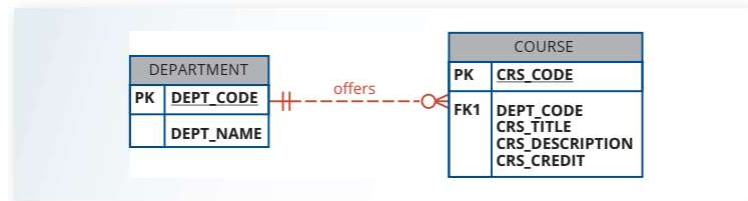


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Tiny College ERD

- ❑ Each department may offer courses.
- ❑ Tiny College had some departments that were classified as “research only,” they would not offer courses
 - The COURSE entity would be optional to the DEPARTMENT entity

Figure 4.28 The Second Tiny College ERD Segment



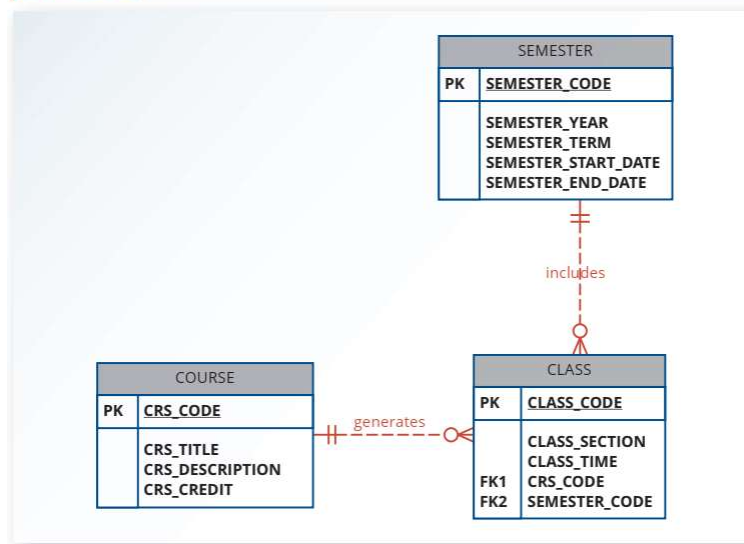
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Tiny College ERD

- A department may offer several sections (classes) of the same course. Each of those classes is taught by a professor at a given time in a given place.
 - In short, a 1:M relationship exists between COURSE and CLASS.
- Each class is offered during a given semester. SEMESTER defines the year and the term that the class will be offered

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Figure 4.29 The Third Tiny College ERD Segment



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Tiny College ERD

- ❑ Each department should have one or more professors assigned to it. One and only one of those professors chairs the department, and no professor is required to accept the chair position.
 - DEPARTMENT is optional to PROFESSOR in the “chairs” relationship.
- ❑ Each professor may teach up to four classes; each class is a section of a course. A professor may also be on a research contract and teach no classes at all.

Figure 4.30 The Fourth Tiny College ERD Segment

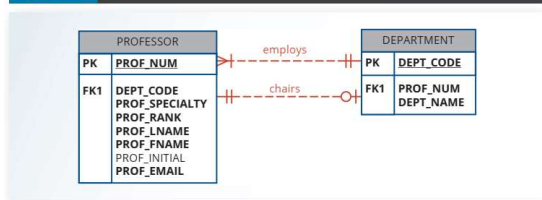
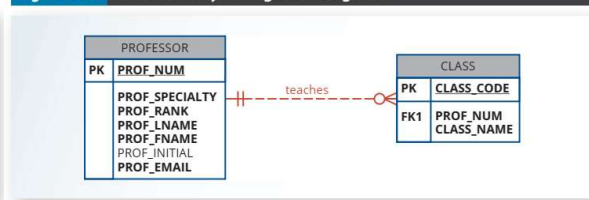


Figure 4.31 The Fifth Tiny College ERD Segment



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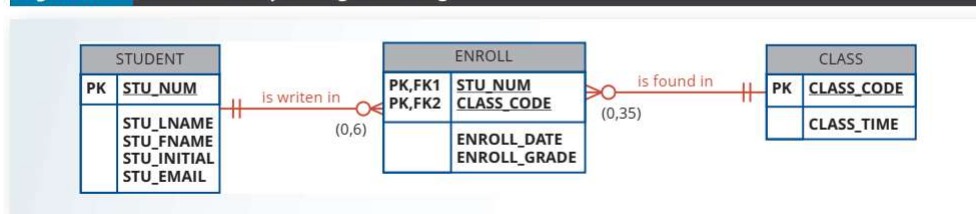
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Tiny College ERD

- ❑ A student may enroll in several classes but take each class only once during any given enrollment period.
- ❑ Each student may enroll in up to six classes, and each class may have up to 35 students

Figure 4.32 The Sixth Tiny College ERD Segment



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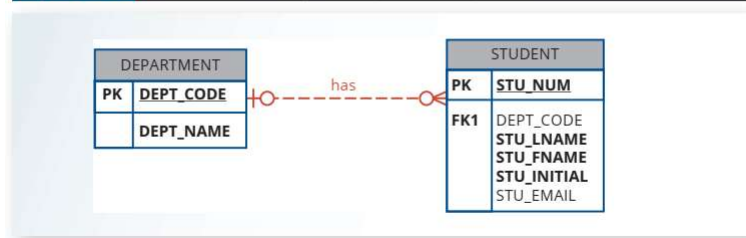
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Tiny College ERD

- ❑ Each department has several (or many) students whose major is offered by that department.
- ❑ Each student has only a single major and is therefore associated with a single department.
 - it is possible for a student not to declare a major field of study

Figure 4.33 The Seventh Tiny College ERD Segment

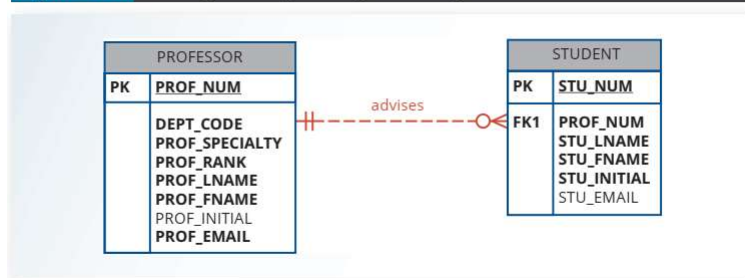


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Tiny College ERD

- ❑ Each student has an advisor in his or her department; each advisor counsels several students.
- ❑ An advisor is also a professor, but not all professors advise students.

Figure 4.34 The Eighth Tiny College ERD Segment

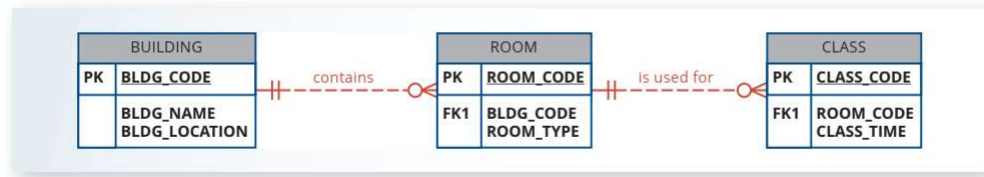


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Tiny College ERD

- A BUILDING can contain many ROOMs, but each ROOM is found in a single BUILDING.
- Some buildings do not contain (class) rooms. For example, a storage building might not contain any named rooms at all.

Figure 4.35 The Ninth Tiny College ERD Segment



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PROFESSOR	SCHOOL	DEPARTMENT
COURSE	CLASS	SEMESTER
STUDENT	BUILDING	ROOM
ENROLL (the associative entity between STUDENT and CLASS)		

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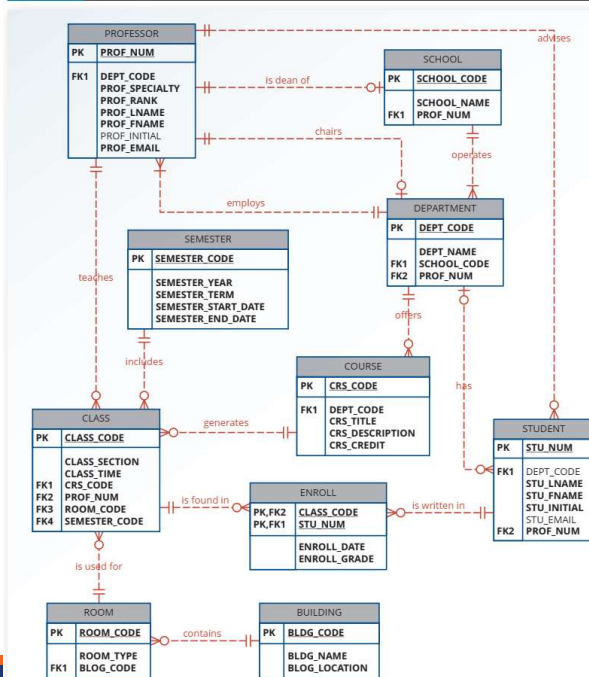
Table 4.4 Components of the ERM

Entity	Relationship	Connectivity	Entity
SCHOOL	operates	1:M	DEPARTMENT
DEPARTMENT	has	1:M	STUDENT
DEPARTMENT	employs	1:M	PROFESSOR
DEPARTMENT	offers	1:M	COURSE
COURSE	generates	1:M	CLASS
SEMESTER	includes	1:M	CLASS
PROFESSOR	is dean of	1:1	SCHOOL
PROFESSOR	chairs	1:1	DEPARTMENT
PROFESSOR	teaches	1:M	CLASS
PROFESSOR	advises	1:M	STUDENT
STUDENT	enrolls in	M:N	CLASS
BUILDING	contains	1:M	ROOM
ROOM	is used for	1:M	CLASS

Note: ENROLL is the composite entity that implements the M:N relationship "STUDENT enrolls in CLASS."

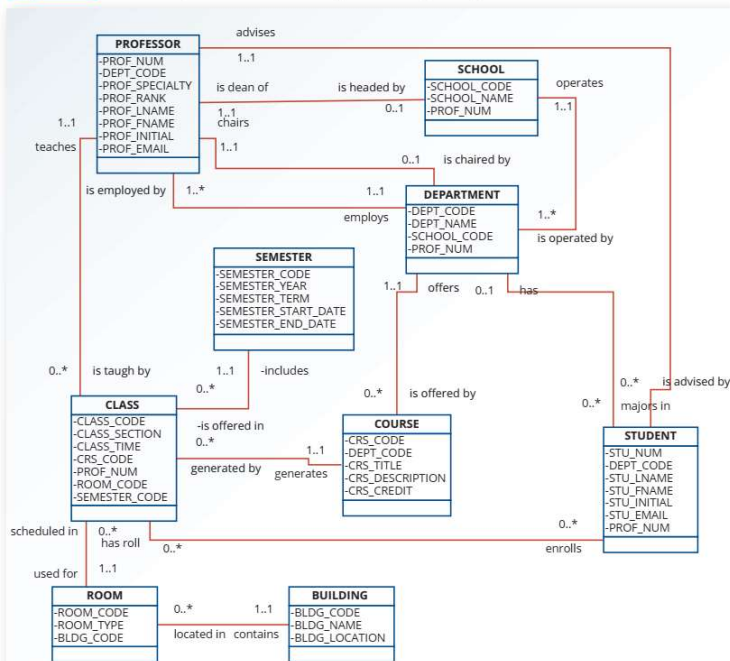
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Figure 4.36 The Completed Tiny College ERD



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Figure 4.37 The Conceptual UML Class Diagram for Tiny College



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4-3 Database Design Challenges: Conflicting Goals

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4-3 Database Design Challenges: Conflicting Goals

- ❑ Conflicting goals: design standards (design elegance), processing speed, and information requirements.
- ❑ *Design standards*: conform to design standards. Design standards govern the presentation of all components within the database design.
 - Minimize data redundancies, minimizing the likelihood that destructive data anomalies will occur. Avoid nulls to the greatest extent possible. ...
- ❑ *Processing speed*. Minimal access time, which may be achieved by minimizing the number and complexity of logically desirable relationships.
 - For example, a “perfect” design might use a 1:1 relationship to avoid nulls, while a design that emphasizes higher transaction speed might combine the two tables to avoid the use of an additional relationship, using dummy entries to avoid the nulls. also be forced to include derived attributes in the design.

4-3 Database Design Challenges: Conflicting Goals

- ❑ Conflicting goals: design standards (design elegance), processing speed, and information requirements.
- ❑ *Information requirements*: Complex information requirements may dictate data transformations, and they may expand the number of entities and attributes within the design. Database may have to sacrifice some of its “clean” design structures and high transaction speed to ensure maximum information generation.
 - A detailed sales report must be generated periodically. The sales report includes all invoice subtotals, taxes, and totals; even the invoice lines include subtotals.
 - The sales report includes hundreds of thousands (or even millions) of invoices, computing the totals, taxes, and subtotals is likely to take some time.
 - If those computations had been made and stored as derived attributes in the INVOICE and LINE tables at the time of the transaction, the real-time transaction speed might have declined

Summary

- The ERM uses ERDs to represent the conceptual database as viewed by the end user. The ERM's main components are entities, relationships, and attributes. The ERD includes connectivity and cardinality notations, and can also show relationship strength, relationship participation (optional or mandatory), and degree of relationship (such as unary, binary, or ternary).
- Connectivity describes the relationship classification (1:1, 1:M, or M:N). Cardinality expresses the specific number of entity occurrences associated with an occurrence of a related entity. Connectivities and cardinalities are usually based on business rules.
- In the ERM, an M:N relationship is valid at the conceptual level. However, when implementing the ERM in a relational database, the M:N relationship must be mapped to a set of 1:M relationships through a composite entity.
- ERDs may be based on many different ERMs. However, regardless of which model is selected, the modeling logic remains the same. Because no ERM can accurately portray all real-world data and action constraints, application software must be used to augment the implementation of at least some of the business rules.
- Unified Modeling Language (UML) class diagrams are used to represent the static data structures in a data model. The symbols used in the UML class and ER diagrams are very similar. The UML class diagrams can be used to depict data models at the conceptual or implementation abstraction levels.
- Database designers, no matter how well they can produce designs that conform to all applicable modeling conventions, are often forced to make design compromises. Those compromises are required when end users have vital transaction-speed and information requirements that prevent the use of "perfect" modeling logic and adherence to all modeling conventions. Therefore, database designers must use their professional judgment to determine how and to what extent the modeling conventions are subject to modification. To ensure that their professional judgments are sound, database designers must have detailed and in-depth knowledge of data-modeling conventions. They must also document the design process from beginning to end, which helps keep the design process on track and allows for easy modifications in the future.